Problem 1: Let V be an inner product space and $\mathbf{u}, \mathbf{v} \in V$. Prove that:

(a) $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\|\mathbf{u}\| + 2\|\mathbf{v}\|$.

(b) $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\|\mathbf{u}+\mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2$

Solution

(a) Use the inner product definition of norm:

$$\|\mathbf{u}+\mathbf{v}\|^2 = \langle \mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$$

$$\|\mathbf{u}-\mathbf{v}\|^2 = \langle \mathbf{u}-\mathbf{v}, \mathbf{u}-\mathbf{v} \rangle = \|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$$

Add both:

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

(b) Subtract:

$$\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\langle \mathbf{u}, \mathbf{v} \rangle \Rightarrow \langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2\right)$$

Problem 2: Let V be an inner product space and $\mathbf{u}, \mathbf{v} \in V$. Prove that:

(a) $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$.

(b) If $\mathbf{u}$ is orthogonal to $\mathbf{v}$ then $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Solution

(a) This is the Cauchy-Schwarz inequality; follow the proof from lectures.

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$$

Equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

(b) If $\mathbf{u} \perp \mathbf{v}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. So:

$$\|\mathbf{u}+\mathbf{v}\|^2 = \langle \mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

$$\boxed{\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2}$$

Problem 3: Let $V = \mathbb{R}^3$ be Euclidean inner product space and $\mathbf{v_1} = (0,1,0), \mathbf{v_2} = (\frac{-4}{5}, 0, \frac{3}{5}), \mathbf{v_3} = (\frac{3}{5}, 0, \frac{4}{5})$ be orthonormal basis for V. Let $\mathbf{u} = (1,1,1) \in V$ with $\mathbf{u} = k_1\mathbf{v_1} + k_2\mathbf{v_2} + k_3\mathbf{v_3}$.

(a) Show that $k_i = \langle \mathbf{u}, \mathbf{v_i} \rangle$ for $i = 1, 2, 3$.

(b) Prove that $\|u\| = \sqrt{{k_1}^2 + {k_2}^2 + {k_3}^2}$

Solution

(a)

$$\langle \mathbf{u}, \mathbf{v_1} \rangle = 0 + 1 + 0 = 1$$

$$\langle \mathbf{u}, \mathbf{v_2} \rangle = -\frac{4}{5} + 0 + \frac{3}{5} = -\frac{1}{5}$$

$$\langle \mathbf{u}, \mathbf{v_3} \rangle = \frac{3}{5} + 0 + \frac{4}{5} = \frac{7}{5}$$

We have

$$1 \cdot \mathbf{v_1} - \frac{1}{5} \cdot \mathbf{v_2} + \frac{7}{5} \cdot \mathbf{v_3}$$

$$= 1 \cdot (0,1,0) - \frac{1}{5} \cdot \left(-\frac{4}{5}, 0, \frac{3}{5}\right) + \frac{7}{5} \cdot \left(\frac{3}{5}, 0, \frac{4}{5}\right)$$

$$= (0,1,0) + \left(\frac{4}{25}, 0, -\frac{3}{25}\right) + \left(\frac{21}{25}, 0, \frac{28}{25}\right)$$

$$= \left(0 + \frac{4}{25} + \frac{21}{25},\ 1 + 0 + 0,\ 0 - \frac{3}{25} + \frac{28}{25}\right) = \left(\frac{25}{25}, 1, \frac{25}{25}\right) = (1,1,1)$$

$$\Rightarrow \mathbf{u} = k_1\mathbf{v_1} + k_2\mathbf{v_2} + k_3\mathbf{v_3} \quad \text{where} \quad k_i = \langle \mathbf{u}, \mathbf{v_i} \rangle \text{ for } i = 1, 2, 3.$$

(b) Since the basis is orthonormal:

$$\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle = \left\langle \sum_{i=1}^{3} k_i \mathbf{v_i}, \sum_{j=1}^{3} k_j \mathbf{v_j} \right\rangle = \sum_{i=1}^{3} k_i^2 \langle \mathbf{v_i}, \mathbf{v_i} \rangle = \sum_{i=1}^{3} k_i^2$$

So:

$$\boxed{\|\mathbf{u}\| = \sqrt{k_1^2 + k_2^2 + k_3^2}}$$

Problem 4:

(a) Let V be an inner product space and $S = \{\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_n}\}$ is an orthogonal set of nonzero vectors in V. Prove that S is linearly independent.

(b) Let $V = \mathbb{R}^2$ be Euclidean inner product space. Apply the Gram–Schmidt process to transform the basis vectors $\mathbf{u_1} = (1, 1), \mathbf{u_2} = (0, 1)$ into an orthonormal basis $\{\mathbf{v_1}, \mathbf{v_2}\}$

Solution

(a) Let $S = \{\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_n}\}$ be an orthogonal set of nonzero vectors in an inner product space V.
Suppose:

$$a_1\mathbf{v_1} + a_2\mathbf{v_2} + \cdots + a_n\mathbf{v_n} = \mathbf{0}$$

Take inner product with $\mathbf{v_k}$ on both sides:

$$\langle a_1\mathbf{v_1} + \cdots + a_k\mathbf{v_k} + \cdots + a_n\mathbf{v_n}, \mathbf{v_k}\rangle = 0 \Rightarrow a_k\|\mathbf{v_k}\|^2 = 0 \Rightarrow a_k = 0 \quad \text{since } \|\mathbf{v_k}\| \neq 0$$

This holds for all $k = 1, \ldots, n$. Hence S is linearly independent.

$$\boxed{S \text{ is linearly independent}}$$

(b) Apply Gram–Schmidt to $\mathbf{u_1} = (1, 1), \mathbf{u_2} = (0, 1)$:
Step 1: Let $\mathbf{v_1} = \frac{\mathbf{u_1}}{\|\mathbf{u_1}\|} = \frac{(1,1)}{\sqrt{2}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
Step 2: Orthogonalize $\mathbf{u_2}$ against $\mathbf{v_1}$:

$$\text{proj}_{\mathbf{v_1}}(\mathbf{u_2}) = \langle \mathbf{u_2}, \mathbf{v_1}\rangle \mathbf{v_1} = \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$\mathbf{w_2} = \mathbf{u_2} - \text{proj}_{\mathbf{v_1}}(\mathbf{u_2}) = (0, 1) - \left(\frac{1}{2}, \frac{1}{2}\right) = \left(-\frac{1}{2}, \frac{1}{2}\right)$$

$$\mathbf{v_2} = \frac{\mathbf{w_2}}{\|\mathbf{w_2}\|} = \frac{1}{\sqrt{(-1/2)^2 + (1/2)^2}}\left(-\frac{1}{2}, \frac{1}{2}\right) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$\boxed{\mathbf{v_1} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \mathbf{v_2} = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)}$$

Problem 5: Let V be an inner product space and $\mathbf{u}, \mathbf{v} \in V$. Prove that:

(a) $\|\mathbf{u}\| \geq 0$

(b) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

(c) $\|k\mathbf{u}\| = |k|\|\mathbf{u}\|$

Solution

(a) By definition of norm from inner product:

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} \geq 0$$

because $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ for all $\mathbf{u} \in V$.

$$\boxed{\|\mathbf{u}\| \geq 0}$$

(b) (Triangle Inequality):

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2 \leq \|\mathbf{u}\|^2 + 2|\langle \mathbf{u}, \mathbf{v} \rangle| + \|\mathbf{v}\|^2$$

Using Cauchy-Schwarz: $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$,

$$\Rightarrow \|\mathbf{u} + \mathbf{v}\|^2 \leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 = (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$$

$$\Rightarrow \|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

$$\boxed{\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|}$$

(c) From the definition:

$$\|k\mathbf{u}\| = \sqrt{\langle k\mathbf{u}, k\mathbf{u} \rangle} = \sqrt{k^2 \langle \mathbf{u}, \mathbf{u} \rangle} = |k|\|\mathbf{u}\|$$

$$\boxed{\|k\mathbf{u}\| = |k|\|\mathbf{u}\|}$$

Problem 6: Let V be an inner product space. Show that if $\mathbf{w}$ is orthogonal to both $\mathbf{u_1}$ and $\mathbf{u_2}$, it is orthogonal to $k_1\mathbf{u_1} + k_2\mathbf{u_2}$ for all scalars k_1 and k_2. Interpret this result geometrically in the case where V is $\mathbb{R}^3$ with the Euclidean inner product.

Solution

Assume $\langle \mathbf{w}, \mathbf{u_1} \rangle = 0$ and $\langle \mathbf{w}, \mathbf{u_2} \rangle = 0$. Let $\mathbf{v} = k_1\mathbf{u_1} + k_2\mathbf{u_2}$, where $k_1, k_2 \in \mathbb{R}$. Then:

$$\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{w}, k_1\mathbf{u_1} + k_2\mathbf{u_2} \rangle = k_1\langle \mathbf{w}, \mathbf{u_1} \rangle + k_2\langle \mathbf{w}, \mathbf{u_2} \rangle = k_1 \cdot 0 + k_2 \cdot 0 = 0$$

Hence, $\mathbf{w}$ is orthogonal to any linear combination of $\mathbf{u_1}$ and $\mathbf{u_2}$.

Geometric Interpretation: In $\mathbb{R}^3$, the set of all linear combinations $k_1\mathbf{u_1} + k_2\mathbf{u_2}$ spans a plane. Since $\mathbf{w}$ is orthogonal to both $\mathbf{u_1}$ and $\mathbf{u_2}$, it is orthogonal to the entire plane they span. Thus, $\mathbf{w}$ is a normal vector to that plane.

Problem 8: Let $\mathbb{P}_2$ have the inner product $\langle p, q \rangle = \int_{-1}^{0} p(x)q(x)dx$. Apply the Gram–Schmidt process to transform the standard basis $\{1, x, x^2\}$ into an orthonormal basis.

Solution

Same as problem 7.

Problem 7: Let $\mathbb{P}_2$ have the inner product $\langle p, q\rangle = \int_0^1 p(x)q(x)dx$. Apply the Gram–Schmidt process to transform the standard basis $\{1, x, x^2\}$ into an orthonormal basis.

Solution

Consider that:

$$\mathbf{u}_1 = 1, \quad \mathbf{u}_2 = x, \quad \mathbf{u}_3 = x^2$$

Step 1: Set $\mathbf{v}_1 = \mathbf{u}_1 = 1$

$$\|\mathbf{v}_1\| = \sqrt{\langle 1, 1\rangle} = \sqrt{\int_0^1 1\,dx} = \sqrt{1} = 1 \Rightarrow \mathbf{e}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = 1$$

Step 2: Compute

$$\mathbf{v}_2 = x - \frac{\langle x, 1\rangle}{\langle 1, 1\rangle} \cdot 1 = x - \int_0^1 x\,dx = x - \frac{1}{2}$$

$$\|\mathbf{v}_2\| = \sqrt{\int_0^1 \left(x - \frac{1}{2}\right)^2 dx} = \sqrt{\int_0^1 x^2 - x + \frac{1}{4}\,dx} = \sqrt{\frac{1}{3} - \frac{1}{2} + \frac{1}{4}} = \sqrt{\frac{1}{12}} =$$

$$\Rightarrow \mathbf{e}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \sqrt{12}\left(x - \frac{1}{2}\right)$$

Step 3: Compute

$$\mathbf{v}_3 = x^2 - \frac{\langle x^2, \mathbf{v}_1\rangle}{\langle \mathbf{v}_1, \mathbf{v}_1\rangle} \cdot \mathbf{v}_1 - \frac{\langle x^2, \mathbf{v}_2\rangle}{\langle \mathbf{v}_2, \mathbf{v}_2\rangle} \cdot \mathbf{v}_2$$

First,

$$\langle x^2, \mathbf{v}_1\rangle = \langle x^2, 1\rangle = \int_0^1 x^2 dx = \frac{1}{3}$$

Next,

$$\langle x^2, \mathbf{v}_2\rangle = \left\langle x^2, \left(x - \frac{1}{2}\right)\right\rangle = \int_0^1 x^2\left(x - \frac{1}{2}\right)dx = \int_0^1 x^3 - \frac{1}{2}x^2 dx = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

$$\Rightarrow \mathbf{v}_3 = x^2 - \frac{1}{3} - \left(x - \frac{1}{2}\right) = x^2 - x + \frac{1}{6}$$

$$\|\mathbf{v}_3\|^2 = \int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx = \int_0^1 \left(x^4 - 2x^3 + \frac{4x^2}{3} - \frac{x}{3} + \frac{1}{36}\right)dx$$

$$= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180} \Rightarrow \|\mathbf{v}_3\| = \frac{1}{\sqrt{180}} \Rightarrow \mathbf{e}_3 = \sqrt{180}\left(x^2 - x + \frac{1}{6}\right)$$

Final Orthonormal Basis:

$$\boxed{\left\{1, \quad \sqrt{12}\left(x - \frac{1}{2}\right), \quad \sqrt{180}\left(x^2 - x + \frac{1}{6}\right)\right\}}$$